

EDUCATION

Master of Science in Computer Science, University of Georgia, CGPA: 3.85 / 4.0 Aug 2023 – May 2025

Relevant Coursework: Data Science, Algorithms, Database Management, Distributed Computing Systems

Bachelor of Technology, Computer Science, GNITS Aug 2018 – Jul 2022

WORK EXPERIENCE

Research Assistant – Software Engineering

Aug 2025 – May 2026

University of Georgia, Remote, USA

- Engineered a **fault-tolerant Kafka** consumer persisting ML prediction results to **PostgreSQL** with **post-persistence manual offset commits, idempotent deduplication, and dead-letter topic routing**, minimizing data loss under transient failures.
- Decoupled email delivery from the signup critical path using **async Kafka** pipeline, **reducing p95 latency by ~350ms**.
- Designed a stateless **JWT-based authentication** and **RBAC authorization** for a **distributed microservices** platform, centralizing token issuance and validation while standardizing authorization across services.
- Built a centralized JWT auth layer with **React Context API** and **Axios interceptors**, consolidating token attachment, refresh, and session handling.
- Instrumented microservices with **Prometheus** and **Grafana** to monitor Kafka consumer lag, p95 latency, and JVM heap.

Graduate Researcher

Aug 2024 – May 2025

University of Georgia, Athens, GA

- Developed an **end-to-end ML pipeline** forecasting alfalfa biomass yields across 4 NY locations using 20 years of agricultural trial data (7,344 records), achieving a best **R² of 0.91** and **SMAPE of 2.99%** via **Optuna-tuned XGBoost**.
- Improved model performance through **Optuna Bayesian hyperparameter search** and ensemble methods (**voting and stacking regressors**), **boosting R² by up to 10.5%** and **cutting SMAPE by up to 11.6%** across four cities.
- Devised a two-stage augmentation pipeline using uniform perturbation and **SMOTE** to transform aggregated agricultural samples into temporally balanced training records for regression modeling.
- Presented *Ensemble Learning for Alfalfa Biomass Yield Forecasting* as an accepted poster at the **2026 AI in Agriculture Conference**, NC State University, Raleigh, NC.

Software Engineer I (Full Stack Developer)

Jan 2022 – Jul 2023

NCR Voyix, Hyderabad, India

- Reduced NCR Counterpoint API latency by **93%** by engineering a **Redis caching layer** with **multi-tenant namespace isolation**, reducing redundant database reads for pay codes, tax codes, and gift card codes on every transaction.
- Integrated **Elasticsearch** as a search read model, enabling **sub-20ms full-text search** with **fuzzy matching** and relevance ranking improving the latency by **10x** over full-table-scans.
- Optimized the customers endpoint using **JPA pagination** and **DTO projections** on lazy-loaded collections, preventing **N+1 queries** and reducing risk of connection pool exhaustion under heavy sync load.
- Built the User Access Control and Role Management modules for the admin console using **TanStack Query** and **Zustand**, with transactional role assignment, zero-role user warnings, and domain-grouped permission picker.
- Authored comprehensive JUnit and Mockito unit tests, establishing an **80% code coverage** guardrail for the deployment pipeline.

Software Engineer Intern

May 2021 – July 2021

S&P Global, Hyderabad, India

- Architected 15+ **Cloud Custodian** policies using **YAML**, **CloudTrail**, and **SQS** across **8 AWS services** (EC2, S3, IAM, DynamoDB, ECS, ECR, Lambda, DMS) to automate compliance enforcement and reduce **security drift by 90%**.
- Implemented tagging, IAM, and ECR governance policies in **Cloud Custodian**, reducing **non-compliant resources by 95%** and improving cost attribution.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, SQL

Frameworks & Technologies: Spring Boot, Spring Security, React, Redux, Apache Kafka, Spring Data JPA, Hibernate, HTML, CSS

Databases: PostgreSQL, MongoDB, MySQL, Oracle, Redis, Elasticsearch

Cloud & DevOps: AWS, Docker, Jenkins, CI/CD Pipelines, Prometheus, Grafana

Testing & Build Tools: JUnit 5, Mockito, Maven, Postman, OpenAPI 3.0 (Swagger)

AI/ML: Scikit-learn, XGBoost, Optuna, Time Series Forecasting, Data Augmentation & Resampling

Developer Tools & Methodologies: Microservices, Event-driven architecture, Git, Jira, Agile, Scrum

Certifications: Microsoft Certified: Azure Fundamentals | AWS Cloud Practitioner | IBM Full Stack Software Developer Professional Certificate